

ZUNAIRA FATIMA

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PROFESSIONAL SUMMARY

Computer Science engineering student with hands-on experience in Python, SQL, data analysis, data pipelines, machine learning, APIs, and AI applications. Experienced in data preprocessing, validation, visualization, and predictive modelling, with additional exposure to Azure AI, Generative AI, language models, and multi-agent AI through technical training. Strong problem-solving and collaborative project experience, seeking to contribute to AI-assisted data engineering, analytics, automation, and responsible AI initiatives.

EDUCATION

M.Tech in Computer Science & Engineering – Stanley College of Engineering and Technology (Autonomous), Osmania University | Pursuing
B.E. in Computer Science & Engineering – Methodist College of Engineering & Technology (Autonomous), Osmania University | 2022–2026
CGPA: 9.60 | Academic Topper (2022–23)

TECHNICAL SKILLS

Programming: Python, SQL, C, C++, Java, JavaScript, PHP

Data & AI: Data Analysis, Data Preprocessing, Data Validation, Machine Learning, NLP, Generative AI, Language Models, Predictive Modelling

Data Engineering & APIs: Data Pipelines, ETL/ELT Concepts, APIs, TFRRecord Pipelines, Data Quality, Data Documentation

ML/Data Libraries: Pandas, NumPy, Scikit-Learn, TensorFlow, PyTorch, Matplotlib, OpenCV, NLTK, XGBoost

Tools & Platforms: Power BI, Tableau, Git, Firebase, Android Studio, Microsoft Azure AI/Cognitive Services

EXPERIENCE

Data Analyst Intern (Power BI) – SmartBridge | September–October 2025

- Collected, cleaned, transformed, and analyzed structured datasets to identify trends, patterns, and business insights.
- Built interactive Power BI dashboards to track KPIs and support data-driven business reporting.
- Applied preprocessing and validation techniques to improve data accuracy, consistency, and documentation.

Front-End Intern – AICTE/Edunet Foundation | August–September 2025

- Completed a 6-week development internship focused on React, Vite, CSS, and JavaScript.
- Worked with API-based application functionality and collaborative software development concepts.

Data Science Intern – CodeClause | May–June 2025

- Performed exploratory data analysis, statistical modelling, and data validation on large datasets.
- Built data pipelines and analytical dashboards using Python, Pandas, Matplotlib, and Power BI.
- Applied data science algorithms to derive actionable insights from structured datasets.

PROJECTS

AI Hybrid Railway Digital Twin Framework – Team Lead – XGBoost, Reinforcement Learning, SUMO |

- Developed an AI-based Digital Twin framework for railway traffic management using XGBoost and Deep Q-Learning.
- Simulated railway operations and optimized train scheduling for mixed passenger and freight traffic.
- Achieved 91.4% scheduling accuracy, with delay prediction performance of MAE 0.33 min and RMSE 0.44 min.
- Enabled real-time decision support to reduce cascading delays and improve operational efficiency on congested railway corridors.

Deepfake Detection System – Team Lead – TensorFlow, MobileNetV2, MTCNN |

- Built a MobileNetV2 + MTCNN deepfake classifier achieving 95% validation accuracy.
- Optimized TFRRecord data pipelines and automated model validation workflows, strengthening repeatability and data-processing reliability.

Global Food Production Analysis – Python, Pandas, Power BI |

- Analyzed 60 years of FAO agricultural data to evaluate yield, regional trends, and food-security metrics.
- Built interactive dashboards for country-wise comparisons and forecasting.

Travel Planner Web App – React, Vite, JavaScript, APIs |

- Integrated API-based location, weather, and activity data to support automated trip-planning workflows and a user-friendly experience.

IoT-Based Water Quality Monitor – IoT, C++ |

- Developed a real-time water-quality device with $\pm 5\%$ TDS accuracy using calibrated sensors.
- Implemented telemetry for system-health and performance tracking.

AI & CLOUD TRAINING

- ICT Academy and IBM Microsoft Azure AI Technical Training (2026): Hands-on work with Azure AI and Cognitive Services, language models, Generative AI, text analytics, OCR, image analysis, and multi-agent AI solutions through live lab sessions.

CERTIFICATIONS

- AI-Enabled Applications for Customer Service – IBM (December 2025)
- Getting Started with Data – IBM (September 2025)
- Cloud Infrastructure 2025 Certified AI Foundations Associate – Oracle (August 2025)

- TATA Data Analytics Virtual Experience Program – Forage (May 2025)
- Data Analysis with Python – freeCodeCamp (March 2025)
- PCAP: Programming Essentials in Python – Cisco (May 2024)
- Artificial Intelligence Fundamentals – IBM (November 2023)

PUBLICATION & ACTIVITIES

- A Review of AI-Driven Railway Traffic Management Using Digital Twin Technology – International Journal of Innovative Research in Engineering and Management (IJIREM), Vol. 13, Issue 2, 2026, pp. 103–107, ISSN 2350-0557, DOI: 10.55524/ijirem.2026.13.2.14.
- Academic Topper (B.E. CSE) – 2022–23.
- Participant, T-Hub Idea Bootcamp – IoT prototype development under tight deadlines.
- Cybersecurity & Ethical Hacking Workshop – hands-on training in security principles.
- ICT Academy Entrepreneurship Development Program (2025) – innovation, leadership, and startup development under industry mentorship.